A High-Radix and Parametric Satellite Communication System

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Title: A High-Radix and Parametric Satellite Communication System

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Cross-Reference to Related Applications

This application is related to the inventions described in the U.S. Patent Applications for "A Method and System for High-Radix Physical Layer Data Encoding and Representation" and "A Protocol Stack and System for Parametric Data Networking." The present invention extends these principles to the unique environment of satellite communication, the contents of which are hereby incorporated by reference in their entirety.

Abstract

A satellite communication system configured for high-radix physical layer encoding and parametric data transport over space-to-Earth and Earth-to-space links. The system comprises ground stations and satellite transponders equipped with high-radix RF transceivers capable of generating and distinguishing N discrete, distinguishable RF states (where N greater than 2) optimized for the specific challenges of satellite channels (e.g., free-space path loss, atmospheric effects, Doppler shift). Data is transmitted as unevaluated parametric expressions, ensuring end-to-end preservation of data precision across the satellite link, from source to

destination, without loss of fidelity due to intermediate binary conversion or processing, and enabling the robust communication of perfect-precision symbolic data.

Background of the Invention

Satellite communication systems are indispensable for global connectivity, broadcasting, and remote sensing. However, like all current communication technologies, they operate on a binary foundation. Data is converted into bit-streams, modulated onto radio frequency (RF) carriers, and transmitted across vast distances. This binary constraint forces high-precision and symbolic data to be approximated into fixed-width formats, leading to inherent loss of fidelity before transmission. Furthermore, the unique challenges of satellite links—such as significant free-space path loss, atmospheric attenuation, signal delays, and Doppler shifts—require complex error correction and compensation techniques, all applied to the underlying binary data.

Summary of the Invention

The present invention introduces a satellite communication system that operates on a high-radix physical layer and a parametric networking protocol, enabling the native, end-to-end transport of symbolic and perfect-precision data over satellite links. This paradigm shift addresses the limitations of binary encoding and the challenges of space-based communication.

The system comprises: (1) **High-Radix Satellite Transceivers:** Ground stations and satellite transponders are equipped with transceivers capable of generating and reliably distinguishing N discrete RF states (N greater than 2). These states are optimized for the specific characteristics of satellite channels, considering factors like propagation through vacuum and atmosphere, link budgets, and Doppler effects. The states directly map to abstract symbols, replacing traditional binary modulation. (2) **Parametric Protocol Stack:** The network utilizes a parametric protocol stack, extending end-to-end symbolic data transport across the space segment. Parametric packets, with unevaluated mathematical expressions as payloads, traverse the satellite link, preserving data precision. (3) **Adaptive Link Management:** The system incorporates mechanisms to compensate for satellite-specific phenomena, such as dynamic adjustment of the symbol-to-state mapping to counteract Doppler shifts, or adaptive power control to maintain distinguishability over varying path losses, all within the high-radix framework.

This architecture allows for the direct transmission of perfect-precision data, such as complex mathematical expressions or scientific constants, to and from orbital assets and remote ground terminals, without degradation. It significantly enhances the utility of satellite communication for applications demanding absolute data fidelity.

Statement of License Grant to the United States Government

The inventor, Reinaldo Javier Aguiar Marcano, hereby grants to the United States Federal Government and its military and defense agencies a perpetual, royalty-free, non-exclusive license to practice and use the invention described herein for any and all governmental, defense, and non-commercial purposes. This license explicitly excludes any and all state and local governments.

Brief Description of the Drawings

FIG. 1 is a conceptual diagram illustrating the high-radix and parametric data flow in a satellite communication system.

FIG. 2 is a block diagram of a satellite transponder or ground station transceiver incorporating a high-radix RF physical layer with adaptive link management.

Detailed Description of the Invention

Referring to **FIG. 1**, a high-radix and parametric satellite communication system (100) is shown. A ground station (110) acts as a source or destination for symbolic data structures (115), which are encapsulated into parametric packets (120). The ground station's high-radix RF transceiver (118) directly modulates the symbols of the packet onto an RF carrier, generating a high-radix RF signal (125). This signal is transmitted via an antenna (122) to a satellite (130) in orbit. The satellite's transponder (132), also equipped with a high-radix RF transceiver (138), receives the signal via its antenna (134), processes it (e.g., for frequency conversion or amplification) in its high-radix symbolic form, and retransmits it as a high-radix RF signal (145) to another ground station (150). This receiving ground station's high-radix RF transceiver (158) demodulates the signal back into the original parametric packet (120), delivering the perfectly precise symbolic data structure (115) to the destination. Adaptive

link management (e.g., Doppler compensation, power control) is performed in the symbolic domain to maintain signal integrity.

FIG. 2 illustrates a block diagram of a transceiver (200) suitable for a ground station or satellite transponder. It comprises a High-Radix RF Modulator (210) which takes abstract symbols as input and generates N discrete RF states on the antenna (220). An Adaptive Link Manager (260) continuously monitors channel conditions (e.g., signal strength, Doppler shift) and provides feedback to optimize the symbol-to-state mapping used by the modulator and the High-Radix RF Demodulator (230). The demodulator receives RF signals from the antenna (220), distinguishes the N states, and outputs the corresponding abstract symbols (240). The adaptive manager ensures that the high-radix states remain reliably distinguishable despite the dynamic and challenging nature of satellite links.

Claims

What is claimed is:

1. A satellite communication system, comprising:

    - a ground station, comprising:
        - a high-radix RF modulator configured to receive a sequence of abstract symbols and generate a corresponding sequence of N discrete, distinguishable radio frequency (RF) states, where N is an integer greater than two;
        - an antenna for transmitting said sequence of RF states towards a satellite; and
    - a satellite transponder, comprising:
        - an antenna for receiving said sequence of RF states;
        - a high-radix RF transceiver configured to process said sequence of RF states in its symbolic form; and
        - an antenna for retransmitting said sequence of RF states towards a receiving ground station;
    - a receiving ground station, comprising:
        - an antenna for receiving said retransmitted sequence of RF states; and
        - a high-radix RF demodulator configured to detect and distinguish the N discrete RF states and generate a corresponding sequence of abstract symbols.

2. The system of claim 1, further comprising an adaptive link manager configured to compensate for satellite channel impairments, including Doppler shift, by dynamically adjusting the definition or mapping of the N RF states.

3. The system of claim 1, wherein the abstract symbols are components of a parametric data packet, the payload of which is an unevaluated mathematical expression.

4. The system of claim 3, wherein the high-radix RF modulator is configured to directly modulate the parametric packet's payload without intermediate binary conversion.

5. The system of claim 1, wherein the high-radix RF transceiver on the satellite transponder processes the data in its symbolic form without converting it to a binary representation.

6. A method for satellite communication, comprising:

   - at a transmitting ground station, receiving a data structure representing an abstract symbol;
   - generating, based on said abstract symbol, one of N discrete, distinguishable radio frequency (RF) states, where N is an integer greater than two;
   - transmitting said RF state towards a satellite;
   - at the satellite, receiving said RF state, processing it symbolically, and retransmitting it towards a receiving ground station;
   - at the receiving ground station, receiving said RF state; and
   - distinguishing said RF state from the other N-1 possible RF states and generating the original abstract symbol.

7. The method of claim 6, further comprising dynamically compensating for Doppler shift by adjusting the interpretation of the N RF states.

8. The method of claim 6, wherein the abstract symbols represent an unevaluated mathematical expression that is transported across the satellite link without loss of precision.

Drawings

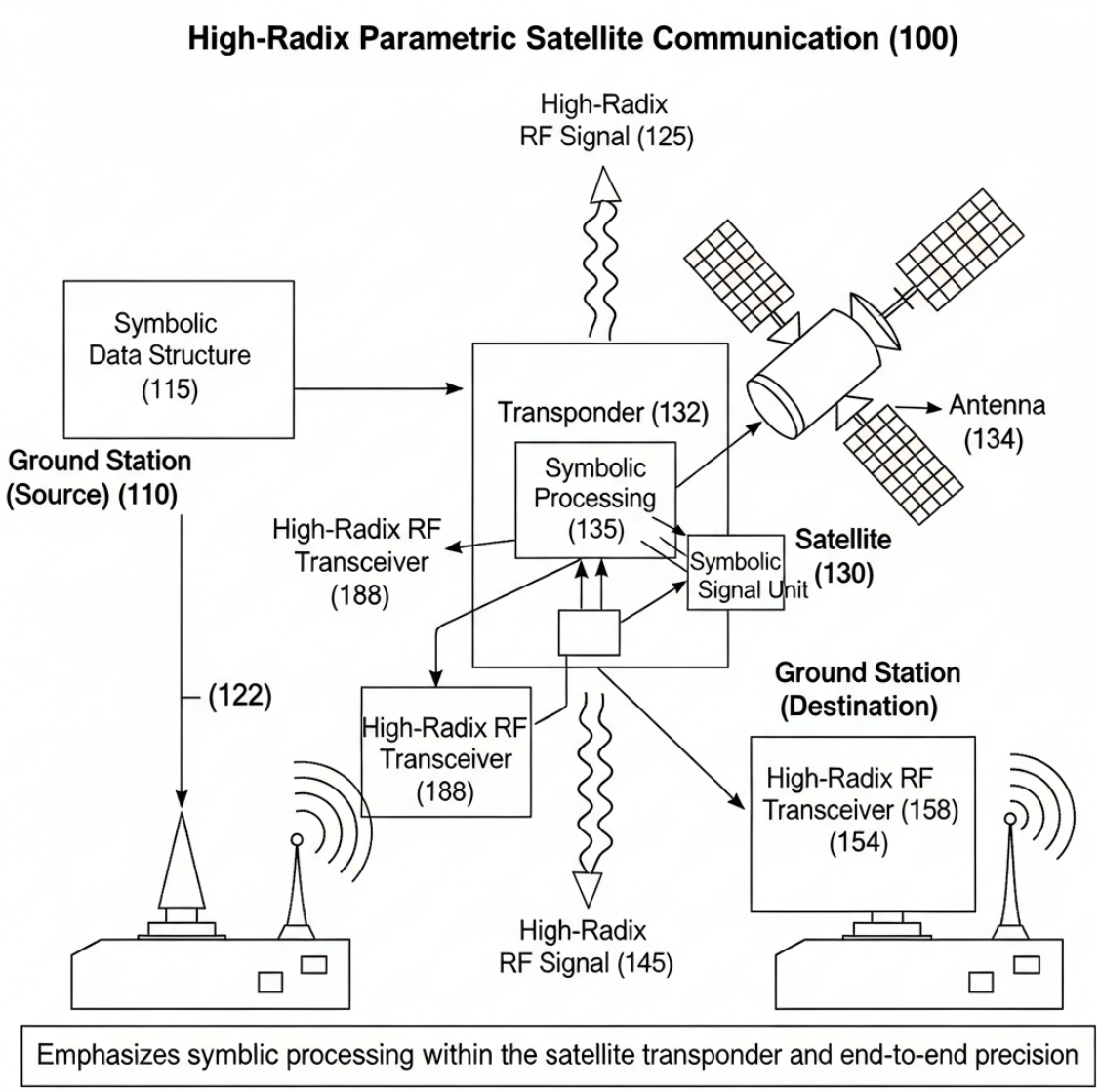


Figure 1: Conceptual diagram illustrating the high-radix and parametric data flow in a satellite communication system.

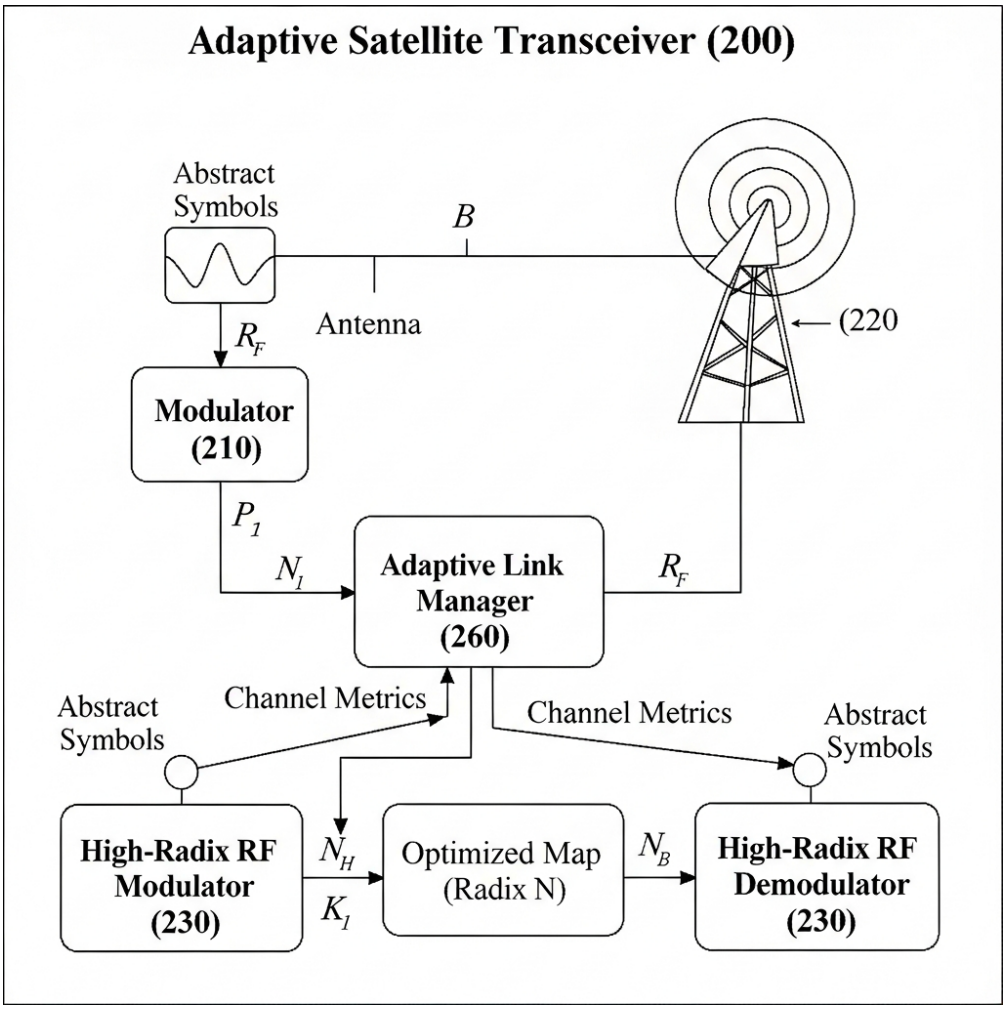


Figure 2: Block diagram of a satellite transponder or ground station transceiver incorporating a high-radix RF physical layer with adaptive link management.